CS502 - Fundamentals of Algorithms Solved MCQs from Quiz#1

CS502 - Fundamentals of Algorithms Solved MCQs from Quiz#1

Solved by Umair Saulat

Heaps can be stored in arrays without using any pointers; this is due to the _____ nature of the binary tree,
left-complete (http://www.vuzs.info/)
right-complete
tree nodes
tree leaves

Sieve Technique can be applied to selection problem?
True (http://www.vuzs.info/)
False

A heap is a left-complete binary tree that conforms to the _____
increasing order only
decreasing order only
heap order (http://www.vuzs.info/)
(log n) order

A (an) _____ is a left-complete binary tree that conforms to the heap order
heap (http://www.vuzs.info/)
binary tree
binary search tree
array

Divide-and-conquer as breaking the problem into a small number of
pivot
Sieve
smaller sub problems (http://www.vuzs.info/)
Selection

In Sieve Technique we do not know which item is of interest
True (http://www.vuzs.info/)
False

The recurrence relation of Tower of Hanoi is given below T(n)={1 if n=1 and 2T(n-1) if n >1 In order to move a tower of 5 rings from one peg to another, how many ring moves are required?
16
10
32 (http://www.vuzs.info/)
31

In the analysis of Selection algorithm, we eliminate a constant fraction of the array with each phase; we get the convergent _____ series in the analysis,
linear
arithmetic
geometric (http://www.vuzs.info/)
exponent

For the heap sort, access to nodes involves simple _____ operations.
arithmetic (http://www.vuzs.info/)
binary
algebraic
logarithmic

For the sieve technique we solve the problem,
recursively (http://www.vuzs.info/)
mathematically
precisely
accurately

The sieve technique works in _____ as follows
phases (http://www.vuzs.info/)
(http://www.vuzs.info/) numbers
integers
routines

Slow sorting algorithms run in,
T(n^2) (http://www.vuzs.info/)
T(n)
T(log n)

A (an) _____ is a left-complete binary tree that conforms to the heap order
heap (http://www.vuzs.info/)
binary tree
binary search tree
array

In the analysis of Selection algorithm, we make a number of passes, in fact it could be as many as,
T(n)
T(n / 2) (http://www.vuzs.info/)
log n
n / 2 + n / 4

The sieve technique is a special case, where the number of sub problems is just
5
many
1 (http://www.vuzs.info/)
few

In which order we can sort?
increasing order only
decreasing order only
increasing order or decreasing order (http://www.vuzs.info/)
both at the same time

Analysis of Selection algorithm ends up with,
T(n)
T(1 / 1 + n)
T(n / 2)
T((n / 2) + n) (http://www.vuzs.info/)

The analysis of Selection algorithm shows the total running time is indeed _____in n,
arithmetic

geometric
linea (http://www.vuzs.info/)r
orthogonal

How many elements do we eliminate in each time for the Analysis of Selection algorithm?
n / 2 elements (http://www.vuzs.info/)
(n / 2) + n elements
n / 4 elements
2 n elements

For the heap sort we store the tree nodes in
level-order traversal (http://www.vuzs.info/)
in-order traversal
pre-order traversal
post-order traversal

One of the clever aspects of heaps is that they can be stored in arrays without using any _____.
pointers (http://www.vuzs.info/)
constants
variables
functions

Divide-and-conquer as breaking the problem into a small number of
pivot
Sieve
smaller sub problems (http://www.vuzs.info/)
Selection

How much time merge sort takes for an array of numbers?
T(n^2)
T(n)
T(log n)
T(n log n) (http://www.vuzs.info/)

The reason for introducing Sieve Technique algorithm is that it illustrates a very important special case of,
divide-and-conquer (http://www.vuzs.info/)
decrease and conquer
greedy nature
2-dimension Maxima

The number of nodes in a complete binary tree of height h is
2^(h+1) – 1 (http://www.vuzs.info/)
2 * (h+1) – 1
2 * (h+1)
((h+1) ^ 2) – 1

Sieve Technique applies to problems where we are interested in finding a single item from a larger set of _____
n items (http://www.vuzs.info/)
phases
pointers
constant

Memorization is?
To store previous results for future use
To avoid this unnecessary repetitions by writing down the results of recursive calls and looking them up again if we need them later (http://www.vuzs.info/)
To make the process accurate
None of the above

Which sorting algorithm is faster
O (n log n)
O n^2
O (n+k) (http://www.vuzs.info/)
O n^3

Quick sort is
Stable & in place
Not stable but in place (http://www.vuzs.info/)
Stable but not in place
Some time stable & some times in place

One example of in place but not stable algorithm is
Merger Sort
Quick Sort (http://www.vuzs.info/)
Continuation Sort
Bubble Sort

In Quick Sort Constants hidden in T(n log n) are
Large
Medium
Small (http://www.vuzs.info/)
Not Known

Continuation sort is suitable to sort the elements in range 1 to k
K is Large
K is not known
K may be small or large
K is small (http://www.vuzs.info/)

In stable sorting algorithm.
If duplicate elements remain in the same relative position after sorting (http://www.vuzs.info/)
One array is used
More than one arrays are required
Duplicating elements not handled

Which may be a stable sort?
Merger
Insertion
Both above
None of the above (http://www.vuzs.info/)

An in place sorting algorithm is one that uses ___ arrays for storage
Two dimensional arrays
More than one array
No Additional Array (http://www.vuzs.info/)
None of the above

We do sorting to,
keep elements in random positions
keep the algorithm run in linear order
keep the algorithm run in (log n) order
keep elements in increasing or decreasing order (http://www.vuzs.info/)

In Sieve Technique we donot know which item is of interest
True (http://www.vuzs.info/)
False

Sorting is one of the few problems where provable _____ bonds exits on how fast we can sort,
upper
lower (http://www.vuzs.info/)

average
log n

The reason for introducing Sieve Technique algorithm is that it illustrates a very important special case of,
divide-and-conquer (http://www.vuzs.info/)
decrease and conquer
greedy nature
2-dimension Maxima

Analysis of Selection algorithm ends up with,
T(n) (http://www.vuzs.info/)
T(1 / 1 + n)
T(n / 2)
T((n / 2) + n)

In in-place sorting algorithm is one that uses arrays for storage :
An additional array
No additioanal array (http://www.vuzs.info/)
Both of above may be true according to algorithm
More than 3 arrays of one dimension.

Which sorting algorithn is faster :
O(n^2)
O(nlogn) (http://www.vuzs.info/)
O(n+k)
O(n^3)

The running time of quick sort depends heavily on the selection of
No of inputs
Arrangement of elements in array
Size o elements
Pivot elements (http://www.vuzs.info/)

Which may be stable sort:
Bubble sort
Insertion sort
Both of above (http://www.vuzs.info/)

For the Sieve Technique we take time
T(nk) (http://www.vuzs.info/)
T(n / 3)
n^2
n/3